

Next generation local area networks

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Many people now consider the local area network (LAN) to which their desktop personal computer (PC) is attached to be more important to them than the telephone network, resulting in a drive to increase both the performance and the reliability of LANs. This paper examines the key drivers for new LAN technologies, summarising LAN developments over previous years. It then looks at possible options for future LANs, which will need to evolve to meet changing traffic patterns brought about by widespread use of intranets and multimedia applications. This encompasses both new technologies such as Gigabit Ethernet, and new architectures such as virtual LANs (VLANs), multi-protocol over ATM (MPOA) and multi-protocol over LANs (MPOL).

1. Introduction

The local area network (LAN) has grown up. Long regarded as the *enfant terrible* of the networking family, the LAN has undergone a difficult adolescence, but is now driving many of the changes seen in networking, as new LAN applications — for example, networked multimedia or the more humble intranet World Wide Web (WWW) site — lead the push for ever-higher wide area network (WAN) bandwidths. The importance of LANs has grown to the extent that many people now regard access to electronic mail and the corporate intranet as more important to them at work than the telephone network. This places fresh challenges on the networking and computer systems — the reliability and availability of LANs and servers now has to be comparable to that of the telephony network. This is not trivial — in traditional telephony the characteristics of customer premises equipment (CPE) have been rigorously defined. However, on a LAN, users have a great degree of freedom in what equipment they connect and in how they use (and, in many cases, abuse) the system, and in the applications they run.

Local area networks are of vital business importance to BT and to its customers in many ways:

- BT is a supplier of LAN solutions, and is increasingly looking to manage totally outsourced networks including LANs, so allowing companies to focus on their core business — understanding trends in this rapidly moving field can give advantage over competitors,
- BT provides LAN interconnect solutions based on technologies such as Frame Relay, SMDS and latterly ATM — the intranet phenomenon extends this, and is

brought about by the use of advanced applications across a well-interconnected network, necessitating the use of high-performance LAN equipment in an ‘access’ role, and exploiting intranets requires a detailed understanding of network, application and server technologies,

- there is customer demand to run new LAN applications across WANs, ranging from document sharing to videoconferencing; thus bandwidth increases in the LAN to stimulate new applications, which in turn raise traffic levels across the WAN — additionally, it is vital that the impact of the new applications on the WAN are well understood, for example, applications such as the Internet phone have very different traffic requirements to the World Wide Web (WWW).

BT is becoming ever more reliant on its internal computer resources — use of the BT intranet is increasingly more widespread. This is much more than an interconnect network — it can be exploited to understand the new ways of working which this technology will bring to customers, and allow BT to proactively meet their needs. LANs, being the ‘access’ layer, constitute the largest part of the network both in terms of volume of equipment and support time.

This paper describes trends in LANs over the next 1—5 years, including the kinds of applications which will be enabled. Implications for the WAN are highlighted.

2. Key drivers and desktop trends

Many factors are driving the development of LANs. The most important of these are detailed below.

- Faster desktop machines

Faster desktop machines are capable of sending data to each other at ever greater rates, for example RAM-to-RAM file transfer rates in excess of 100 Mbit/s have been measured on high-end Unix machines across the BT Laboratories (BTL) futures test bed [1]. Several years ago the PC itself was the bottle-neck — processor power and bus speeds negated the need for high-speed LAN technology, except as a backbone, aggregating many traffic flows. With new generations of desktop machines, the network can easily be the bottle-neck. Processor speeds in excess of 200 MHz are now commonplace, while faster buses within PCs have removed the bottle-necks. Faster processors reduce the need for specialist decompression hardware (e.g. for MPEG video), easing the introduction of new applications.

- WWW technology

The use of WWW technology to distribute documents — a common application of intranets — is enabling any desktop machine to talk to any server, irrespective of the operating systems involved, hiding many of the complexities apparent in traditional file-transfer techniques from the user. Intranets are fundamentally changing the traffic patterns in LANs. In the past, 80% of traffic has remained local; in the future it is likely that only 20% will remain local. This is summarised in Fig 1.

In early LANs, typically 80% of traffic remained local to the LAN, with 20% destined outside the LAN. The introduction of the WWW is inverting this relationship, and the ability to forward efficiently outside the LAN is of paramount importance.

- Multimedia applications

A range of non-networked and networked multimedia applications are now available. Non-networked applications, such as games or CD-ROMs, require sound and video cards to be installed. Networked applications require these, plus a network connection. Such applications typically use video and audio delivered over IP. Coupled with 'intranet' growth, IP is becoming a 'common bearer' protocol, allowing a diverse range of applications to be supported from a single protocol stack.

Real-time IP applications such as Internet telephony or the Mbone [2, 3] show that even on current networks, which do not guarantee packet delivery, useful quality can be obtained with comparatively low bandwidths. Collaborative working, such as joint editing of documents or the use of a shared whiteboard, will also be important multimedia applications. These operate better at higher bandwidths where delays can be reduced.

Download applications, such as video- or audio-on-demand, do not require real-time encoding and decoding (these can buffer material to compensate for jitter or packet loss within the network). Examples include BT City Media Network Services [4], which uses MPEG1 delivery of video at around 2 Mbit/s, to Real Audio [5] which delivers AM radio-quality sound at bit rates down to 10 kbit/s.

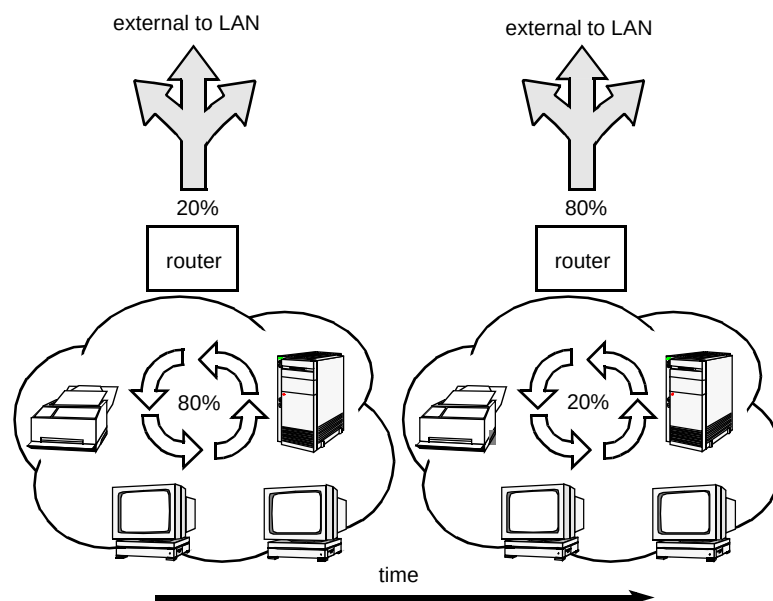


Fig 1 Use of WWW in intranets is fundamentally affecting LAN traffic profiles.

'Traditional' networked applications such as FTP and electronic mail are described as 'elastic' — the user is not concerned about the arrival times of any particular packet, but only the arrival of the whole file within an acceptable period. Real-time applications place new criteria on networks — control of delay is often more important than ensuring the arrival of all packets across the network. These differences are shown in Fig 2, which stresses that video or audio retrieval can be performed in a manner very similar to that used in file transfer, provided the PC has sufficient play-out buffering (> 100 ms) to cope with variations in packet arrival times, and that the network has enough bandwidth to deliver packets at a sufficient rate.

- Network computing

Network computers (NCs) operate by downloading much of their resources from servers. End-to-end server availability becomes even more critical if and when these are widely deployed.

- Support costs

Desktop support can be a large part of the costs of running a LAN — typically 50 PCs are supported by each member of support staff, and, over a typical three-year life cycle of a desktop machine, support costs can be significantly higher than the initial hardware costs. Investment in network infrastructure can thus be justified if it can reduce support costs. The drive is to move to systems which enable more users to be supported by the same number of support staff, while improving the performance of the network.

- Protocol compatibility

Currently IP, IPX and Appletalk protocols dominate the desktop market. The Internet and intranets are facilitating the growth of IP as the default protocol. However, IPX looks likely to be deployed in sizeable numbers for the foreseeable future (for compatibility

with existing Novell Netware systems), and Appletalk will continue to be important, especially in the US and in market segments where it currently has a high take-up (such as publishing). A likely scenario for the medium term is that a significant proportion of desktop machines will run parallel networking stacks, IP for 'Internet' applications, and IPX or Appletalk for connection to local file servers. Networks must be designed to cope with all three of these protocols.

- Network technology

A wide range of network technology choices are becoming available to support the variety of applications described above — LAN switching, 10baseT, 100baseT, Gigabit Ethernet, and ATM. The paper describes the likely roles of each of these technologies in corporate LANs over the next few years.

These trends are summarised in Fig 3, which shows how new LAN technologies play a vital part in stimulating demand for WAN bandwidth, and new applications.

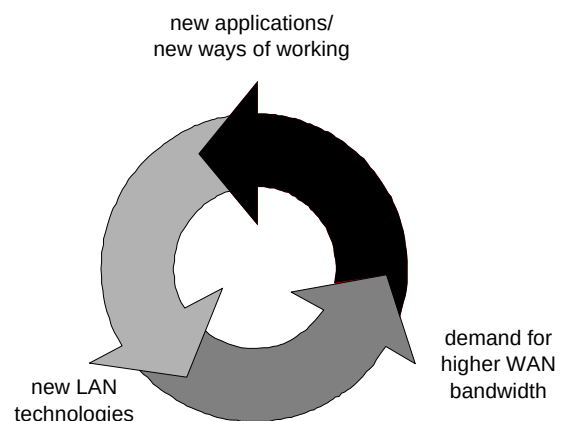


Fig 3 New LAN technologies play a key role in facilitating new applications and driving the demand for WAN bandwidth.

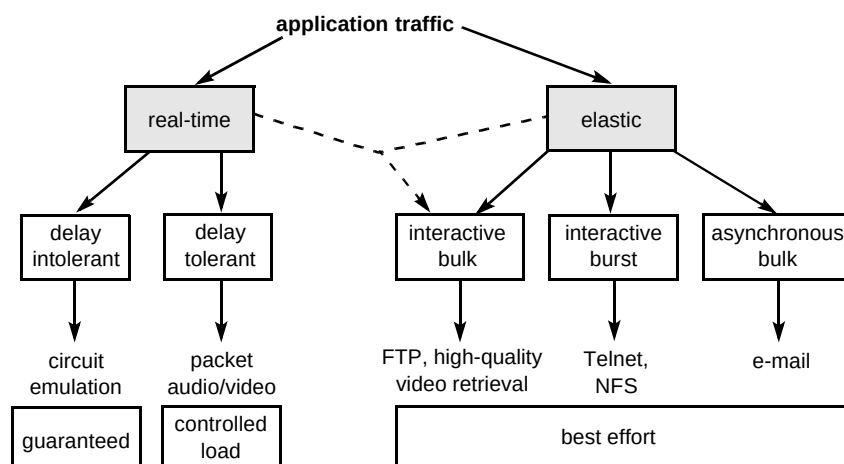


Fig 2 The different classes of application and the resultant traffic types.

3. LAN technologies

A range of technologies are available to the user — 10-Mbit/s Ethernet, 100-Mbit/s Ethernet, Gigabit Ethernet, FDDI, ATM and Token Ring. All have a role in delivery to the desktop or in the backbone, and all will be present to some degree in the future, if only for reasons of legacy. These will be described in this section, along with a discussion of terms such as bridging, routing and LAN switching.

3.1 Ethernet

Ethernet was invented at Xerox's Palo Alto Research Center [6] and formalised as the IEEE802.3 standard by the Institute of Electrical and Electronics Engineers. It uses carrier sense multiple access with collision detection (CSMA/CD). Before transmission of data, stations sense whether the network is already in use. If it is, the station waits before commencing transmission. Collisions occur when multiple stations listen for traffic, detect none and begin transmission at the same time. A back-off mechanism is used, whereby clients wait a random interval in the event of collisions before retransmitting data. Such etiquette gives poor performance when the network has many users and traffic levels are high. Within this standard many different types of wiring may be used including shielded coaxial cable, thin coaxial cable, and unshielded twisted pair or optical fibre.

Ethernet network topology evolution

Traditionally Ethernet has been a shared medium, meaning that many users have shared the same bandwidth. The bus topology typically consisted of a single cable, e.g. thin-wire Ethernet (the familiar piece of co-axial cable running round an office) to which all the users were directly attached in series. Such a system is potentially unreliable since the users have direct access to the physical medium. Since operation depends on transmission line characteristics of the cable, a single user can destroy network performance for all LAN users, for example, by incorrectly terminating the cable. This led to the concept of 'collapsing the backbone' — locking the vulnerable parts of the network away from the users in equipment called a hub, which repeats traffic down all the connected cables. This means moving from a bus topology to a star topology (see Fig 4).

The hub can readily be replaced by a LAN switch (see below), giving a simple upgrade path. An Ethernet thin-wire network is a bus topology, while an Ethernet 10baseT network is the star topology. This is usually associated with 'structured wiring', typically category 5 (or in the USA category 3 which took a large market share) and is known as 10BaseT. Category 5 wiring consists of 4 pairs of twisted wires carried in the same sheath. The variants are UTP (unshielded twisted pair) and STP (shielded twisted pair), of

which the former is more common on cost grounds. Although comparatively inexpensive it is capable of supporting data rates in excess of 100 Mbit/s over a distance of 100 metres. Moving to a collapsed backbone improves reliability by removing the vulnerable physical connections to a secure location, simplifying fault finding. The physical termination of category 5 cabling is on an 8-pin RJ-45 socket. Ethernet can also be supported over optical fibre using the 10BaseF specification.

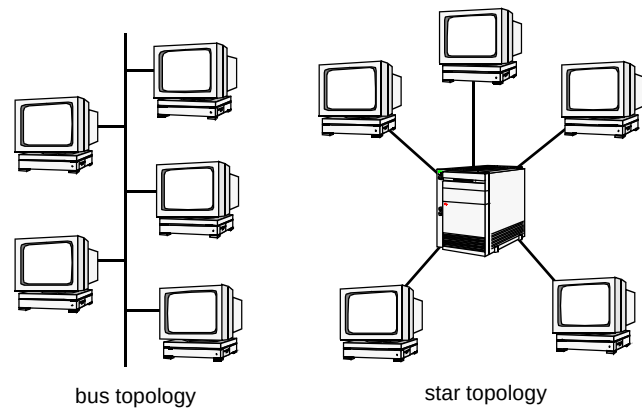


Fig 4 Moving to a star topology by 'collapsing the backbone' through the use of structured wiring.

Segmenting the LAN — bridging and routing

The performance of a hubbed network is still fundamentally that of a shared medium, as all PCs are still subject to all network traffic. With many PCs attached, the need to segment the network becomes apparent, for example, low performance PCs running IPX can crash if they lose contact with their server during a file transfer, as might happen if a pair of Unix workstations are involved in a high-performance file transfer. On a 10 Mbit/s shared Ethernet segment a goodput¹ of around 2—3 Mbit/s is achievable in practice, i.e. around 30% of the theoretical figure. However, TCP (the transmission control protocol used by IP for reliable delivery) has a back-off mechanism which reduces the rate at which it sends in the event of congestion; real-time UDP traffic does not have a similar mechanism and can severely degrade TCP performance on the same network.

Splitting the network into multiple Ethernet segments can overcome this limitation. Such segmentation was first achieved by bridges, i.e. devices with intelligence to forward traffic to another segment only when required. A bridge operates by recording which MAC (media access control) addresses (a unique identifier burnt into the configuration of the network interface card (NIC) of the PC during manufacture) are resident on hosts on either side of the bridge.

¹ Goodput is the amount of useful, i.e. collision-free, traffic which flows. Throughput is the total traffic which flows — in Ethernet networks throughput can be much higher than goodput.

PCs and networking equipment use broadcasts (packets which are forwarded to all machines) and in practice it is this traffic which limits the size of a bridged network to several hundred PCs. A different method is required if the network is to scale beyond this point.

The broadcast limitation was overcome by the introduction of hierarchical network addressing schemes, at layer 3 of the OSI stack, which provide the ability for global networking — potentially allowing any PC to communicate with any other computer anywhere else in the world. The segmentation to hierarchical networks is performed by routers, which act on network addresses. Protocols such as IP, IPX and Appletalk have logical addressing structures, allowing hierarchical networks to be implemented. Each network address must be unique throughout the network ².

Network addressing must be correctly configured for PCs to access network resources. For IP this consists of a network address, subnet mask (which explains how the address space is subdivided) and default gateway (the network address of the closest router). Rather than each PC knowing how to reach every remote location, the router stores the routes to those locations and directs packets out of the appropriate interface. This is a software-intensive function, making routers more expensive than bridges. Additionally, higher layer protocols also permit interworking between different physical media types, for example, allowing a Token Ring-attached PC to communicate to an Ethernet-attached PC across a WAN serial link; such flexibility has helped the growth of the Internet.

LAN switching

The recent increase in power of integrated circuits has introduced the possibility of a LAN switch — this is most easily regarded as a multi-port bridge, with each port connected to one (or a few) users. A network can be highly segmented, with traffic only going to machines for which it is intended. This overcomes the CSMA/CD limitation of Ethernet, effectively giving each end-station dedicated line-rate network access. The move from a hubbed network to switched technology can be relatively simple if structured wiring is present; in particular, PCs need not be affected as network addressing does not have to be reconfigured. However, the level of broadcast traffic will still limit the size of a switched network.

Two distinct architectures are used in LAN switches, namely store-and-forward and cut-through:

- store-and-forward switches buffer each frame and perform error checking prior to forwarding to the egress port,

² There are minor exceptions — private IP addresses allocate a small range of address space that can be reused many times within the core network, but which must be translated at some point to a globally unique address in order to communicate with the Internet.

- conversely, cut-through switches start forwarding the packet as soon as the necessary MAC header destination information has been obtained to determine the appropriate egress port — cut-through can give higher throughput as delays are minimised, but performance can be severely degraded on lines with significant error rates; some LAN switches have the ability to automatically change between modes of operation if the error rate rises above a threshold value.

New Ethernet technologies

10 Mbit/s Ethernet has the largest market share — Dataquest data showed that in 1995 87% of LAN ports shipped (by number) were 10baseT Ethernet. It is the most widely deployed network technology down to the desktop, and these economies of scale have driven down NIC and component costs. As often occurs in the computer industry, economies of scale reduce prices, which in turn lead to more sales and further economies of scale. Migration from hubbed to switched Ethernet gives an easy upgrade route, improving performance without the need to replace network interface cards on existing desktop machines, meaning a minimal learning curve for both users and network managers.

100baseT uses the same CSMA/CD principles, and is a faster version of 10baseT. Costs of 100baseT network interface cards (NICs), hubs and switches are continuing to drop rapidly. Currently the performance offered by 100baseT is warranted only by servers and high-end PCs. Existing Ethernet switches can offer both 10baseT and 100baseT ports — allowing servers to be connected at high speed, and clients with lower speed access. One important point is that 100baseT has the same pin-out and behaviour (admittedly ten times faster) as 10baseT, making migration from 10baseT to 100baseT straightforward.

100baseT over fibre, with a range of 2 km, is likely to become the budget campus-area backboning technology in the short term, interconnecting Ethernet switches. Although 10baseT can be a full duplex technology as it has separate pairs of wires for receive and transmit, the vast majority of NICs have been manufactured to be half duplex. This means that incoming and outgoing packets can collide, reducing peak throughput by typically a factor of three. This oversight was solved for 100baseT, which can be implemented as full-duplex at initial installation. This is particularly important for real-time applications such as videoconferencing, where full duplex operation can prevent collisions in the 'access' links.

Another 100 Mbit/s Ethernet standard, 100 VG-any-LAN³ (IEEE802.12), uses a pin-out and electrical transport that is different to 100baseT, making it more useful in electrically noisy environments, such as on the factory floor. It can be supported over 4-pair, category 3 wiring (or

³ VG refers to voice grade

greater). It uses a demand priority system (a method with some similarities to Token Ring), which gives two priority levels, to enable some quality of service (QoS) support for real-time applications. However, take-up of this technology has been slower than 100baseT, meaning that 100 VG-any-LAN prices have remained higher than 100baseT.

IsoEthernet, as defined in IEEE802.9a, can provide multimedia support by adding a 6.144-Mbit/s isochronous bandwidth to the 10 Mbit/s of Ethernet bandwidth for a total of about 16 Mbit/s. The isochronous connection is organised as a wideband ISDN circuit, consisting of 96 bearer (64 kbit/s) channels. The isochronous bandwidth is circuit switched, giving inherent quality of service. This technology, however, lost out in the market-place to comparatively inexpensive 100baseT technology.

A new generation of Ethernet technology is being developed, Gigabit Ethernet, which is being standardised as IEEE802.3z and was completed in the latter half of 1997. This will allow half and full duplex operations at speeds of 1000 Mbit/s using the 802.3 frame format and the CSMA/CD access method with support for one repeater per collision domain.

Gigabit Ethernet will be supported over a variety of physical media — multimode fibre-optic link over 500 m, a single-mode fibre-optic link with a maximum length of 2 km, and a copper-based (coaxial cable, for example) link over at least 25 m. Gigabit Ethernet will only work over short distances on UTP wiring; while this might preclude widespread delivery to the desktop using Gigabit Ethernet, it need not be a problem for network servers which are increasingly collocated with networking equipment in a server farm. Gigabit Ethernet is likely to become a low-cost method of high-performance backboning of Ethernet switches over fibre. Currently 10baseT Ethernet switches are often supplied with 100baseT ports for uplinking or direct connection to servers. Future 100baseT Ethernet switches will be supplied with Gigabit Ethernet ports to supply similar functionality. A clear migration path can be seen — 100baseT will migrate to Gigabit Ethernet, with 10baseT desktop connections migrating to 100baseT as required. Initial Gigabit Ethernet offerings are supporting the draft 802.1p (see section 3.3), enabling QoS support across the LAN.

A number of proprietary schemes have been proposed to give enhanced backbone performance by using parallel 100baseT links; these may be useful until Gigabit Ethernet becomes widely available, but are likely to only operate between equipment from a single vendor.

Wireless LAN technologies already exist, and seem likely to find wider deployment in the future; more details can be found in Fenton et al [7]. Installation of wiring requires a capital outlay and in some locations is not warranted or is impractical, e.g. stock-taking in a warehouse

with an on-line link to a server, or a short-lived trade-show. Alternatively, locations which are passed through by transit users — such as airport lounges — are good candidates for wireless technology. The technology used typically tries to emulate hard-wired Ethernet (historically, Ethernet itself has its roots in the wireless Aloha system), by using spread-spectrum radio. Radio bandwidth is highly regulated, and only a few frequency bands will be available for this purpose, such as 2.4 or 18 GHz in the UK. Infra-red methods can circumvent the regulatory problem, but there are perceived safety problems with high levels of coherent infra-red radiation. The higher carrier frequency of infra-red gives the possibility of larger data bit rates.

3.2 Fibre distributed data interface (FDDI)

FDDI was the first 100-megabit technology to be used in LANs, and has found a role in backboning campus networks. FDDI operates by using a 100-Mbit/s, token-passing, dual-ring LAN and predominantly uses the fibre-optic transmission medium, although CDDI (a copper twisted-pair variant) also exists. It defines the physical layer and media-access portion of the link layer, and so is roughly analogous to IEEE802.3 and IEEE802.5 in its relationship to the open system interconnection (OSI) reference model. Although it operates at faster speeds, FDDI is similar in many ways to Token Ring. The two networks share many features, including topology (ring), media-access technique (token passing) and reliability features (redundant rings, for example). FDDI has inherent security capabilities, which mean that snooping will be easily detected.

However, FDDI may well be marginalised by Ethernet (100baseT/Gigabit) or high-rate ATM. Switched FDDI can be used to improve backbone performance of existing FDDI networks.

3.3 Enhancements in bridged LANs — IEEE802.1p

IEEE802.1p [8] adds class of service (CoS) capability to 802.x LANs. Three bits specify the user priority for a given frame, giving eight different service classes. The minimal implementation of these services is a strict priority with larger values getting a greater priority (0 = lowest, 7 = highest). LAN switches could use different buffers for different classes of traffic. It is possible to limit the number of queues by combining multiple priorities into the same queue. While this is the minimal interpretation, it is also possible to assign specific behaviours to the user priorities. These are still under study but are likely to evolve to match the needs of applications. 802.1p can be exploited by LAN switches to give some degree of quality of service support between LAN switches, allowing end-to-end QoS across a campus network. This is at least a year away from standardisation and at this stage it is unclear how successful this technique will be.

3.4 ATM

In data networks, QoS refers to the ability of a network to forward data across the network with negligible (controlled) loss and with predictable delay — it is not explicitly a measure of network reliability. Ethernet-based technologies are able to guarantee QoS by over-provisioning bandwidth and removing contention, i.e. ultimately a single user per full-duplex segment. While this can be readily achieved using switched Ethernet, effectively allowing wire-speed connectivity between two machines on the same switch, this has two limitations:

- firstly, this does not allow multiple applications (e.g. multiple video feeds) on the same PC to have individual QoS requirements,
- secondly, it does nothing to solve QoS over the wide-area — even between Ethernet switches.

QoS between routers can be given for IP traffic by the resource reservation protocol (RSVP), which can guarantee each IP stream a certain throughput or delay. The management and tariffing issues of RSVP still need to be resolved. QoS between the Ethernet switch to which an end-station is attached and the closest router is being investigated by IEEE802.1p, but commercial versions are still lacking.

While other LAN-based technologies have tried to solve QoS as a bolt-on, ATM was designed with QoS in mind from the outset. ATM's traffic classes are discussed in Adams and Smith [9].

A complication arises when ATM carries data traffic — such traffic is in variable length packets, ranging up to 1518 bytes on Ethernet. All but the shortest packets must be segmented across many ATM cells. This is usually achieved by AAL5 SNAP (sub-network access protocol) encapsulation, which 'wraps' the datagram in a header and trailer (adding 16 bytes), before segmentation into cells. If an ATM network is not adequately dimensioned, cells may be discarded in times of congestion. A datagram which loses a cell is rendered useless, although techniques to discard the remainder of corrupt datagrams can alleviate these effects to some degree. Early packet discard, for example, involves dropping the remainder of an AAL5 frame if a single cell is dropped. Even so, it is important to dimension ATM networks properly to avoid 'congestive collapse' — when cells from many datagrams are discarded the 'goodput' (i.e. the useful traffic) can be severely degraded. Higher layer protocols such as TCP may cause retransmission, aggravating matters.

ATM technology is now readily available at 622-Mbit/s (STM-4) rates, with 2.4-Gbit/s interfaces under development. ATM was designed to use the transmission

capabilities offered by SDH and SONET, meaning that ATM can be used over inter-continental distances.

One of the original drivers for ATM was for a single technology to span between the LAN and the WAN. While this is true at one level, LAN and WAN ATM have somewhat different properties. In a LAN, bandwidth is governed by the cost of infrastructure installation (e.g. blowing fibre into prepared ducting), meaning that there is minimal incentive to be efficient with LAN utilisation. The reverse is true over the WAN, where repeaters and underlying PDH or SDH bandwidth are used. Traffic shaping and policing ensure that traffic fits the agreed parameters and is thus crucial across the WAN. This means that WAN ATM networks operate at a higher utilisation than LAN ATM networks. While the ATM unassigned bit rate (UBR) service category may be successful across LANs, its usefulness over the WAN is yet to be established. Available bit rate (ABR) and the recently proposed guaranteed frame rate (GFR) [9] may be attractive for typical user traffic, e.g. file transfer, over WANs. GFR assures a user that a guaranteed minimum rate of frames is to be forwarded, and offers a fair share of any remaining bandwidth.

One of the factors which has slowed the predicted uptake of ATM in the LAN is the rate at which switched 10baseT and 100baseT technology have come to market, giving good performance at low cost. Additionally, before LAN emulation (LANE, see section 3.7), direct ATM connections were more complex to configure and manage than their Ethernet equivalents.

3.5 Token Ring

Token Ring is second only to Ethernet in the cumulative number of desktops it serves. It has a high take-up in large, corporate customers, especially in the finance sector. Token Ring was initiated as an IBM-proprietary technique, but was later standardised by the IEEE as IEEE802.5. It can be operated at 4 or 16 Mbit/s, across UTP, STP or fibre networks.

Token Ring uses the concept of token passing. A token is a 3-byte frame, possession of which grants a station the right to transmit data. If an end-station has no data to pass then the token is transferred to the next station. Information is removed when it returns to the source station. Unlike Ethernet, collisions cannot occur, because of the determinism introduced by token operation. This gives a maximum interval before transmission is allowed, which can prevent protocols timing out.

The 'ring' has traditionally been collapsed down to a central point, known as the media access unit (MAU). Intelligence in this equipment can sense cable breaks, and incorrectly operating PCs, providing some degree of fault

tolerance. Embedded ring management allows the LAN to be actively managed.

3.6 Virtual LANs (VLANs)

Traditional LANs have a one-to-one relationship between the logical and physical network. For example, an IP subnet may map on to a single Ethernet segment. A VLAN abstracts the logical from the restrictions of the physical. A VLAN may describe an IP subnet, a workgroup or some other logical grouping. In other words, a VLAN is, at its most basic, a software-defined LAN. A VLAN aims to provide the benefits of broadcast containment, security and simplified administration.

Broadcast traffic can be restricted to a VLAN rather than a physical network segment, so the traffic only goes to those end-systems that may be relevant. Membership of the VLAN can be restricted to certain end-systems, providing a logical partition between workgroups. However, certain end-systems could be members of two (or more) VLANs without causing the VLANs to merge. With traditional LAN-based networks end-system moves and changes would require manual configuration of the end-system and associated routers. With VLANs an end-system can move to a new physical location but retain its existing connectivity.

Membership of a VLAN can be based on physical port, MAC address, protocol, subnet or any combination of these (so-called filter based). Since VLANs are a logical concept, they introduce the possibility of bridging between different physical media. For example, LANE (see section 3.7) allows bridging between Ethernet-attached and ATM-attached machines.

3.7 Multi-protocol over ATM and LAN emulation

LANE Version 1.0

The ATM Forum LAN emulation (LANE) [10–12] standard provides a simple way to use ATM to connect network devices so that they appear to each other as devices on an IEEE802-style network (i.e. Ethernet or Token Ring). It hides the mechanisms of ATM from the MAC layer and higher protocols allowing for rapid adoption of ATM. LANE provides services for address resolution, broadcast, and multicast so that a LAN emulation client (LEC) behaves as it would on a traditional multi-access LAN. A LEC can be either an ATM-attached device (such as a workstation) or a legacy LAN device (such as an Ethernet switch) with an ATM uplink. In the first case the workstation applications (including the IP stack) thinks it is talking to an Ethernet (or Token Ring) LAN. In the latter case the LEC acts as a proxy for the legacy end-systems attached to it, forwarding packets over ATM.

LECs set up ATM virtual circuits (called data direct VCCs) between each other in order to transfer data. VCCs are set up in response to LANE address resolution protocol (ARP) requests (LE_ARPs). When one LEC wants to transmit data to an end-station, it sends an LE_ARP request to discover the destination ATM address for that MAC address. The response may come from the end-station's LEC (if ATM-attached) or a proxy LEC. The source LEC then sets up a data direct VCC to the destination LEC. MAC addresses are maintained in the native format of the type of LAN being emulated (currently only Ethernet and Token Ring). LANE version 1 only supports UBR ATM traffic connections and hence does not exploit the QoS capabilities of ATM.

Traffic between LECs on different VLANs must be routed using an external router; LANE does not provide any layer 3 routing.

LANE Version 2.0

LAN Emulation 2.0 [12] provides enhanced capabilities over version 1.0, including enhanced multicast support, server redundancy, and QoS support. These improve the scalability of VLANs. LANE 2.0 has two parts — LANE user-to-network interface (LUNI) 2.0 and LANE network-to-network interface (LNNI) 2.0.

The ATM Forum has defined a way of implementing VLANs over ATM known as emulated LANs. LUNI 2.0 allows LECs to multiplex certain traffic from multiple ELANs over the same virtual channel, thereby reducing the number of virtual channels needed. LUNI 2.0 also improves QoS support because the traffic-service type can be specified (and hence does not have to be UBR). This will allow higher layers to request a QoS type for a particular destination (within the ELAN). Basic LANE QoS features can exploit the IEEE802.1p model. Each of the eight levels would be associated with its own set of UNI call set-up parameters and each LEC is configured with up to eight sets of call set-up parameters. The call set-up parameters will be defined or recommended by an IETF working group as part of the mapping of the IETF integrated services models to ATM and IEEE802 networks — but this will not prohibit other mappings. LANE 2.0 allows higher layers to access the full connection-oriented QoS characteristics of the underlying ATM network, which could be based on user-network interface (UNI) 4.0.

LUNI 2.0 also adds support for separating out multicast traffic so that only those end-systems in that multicast group will receive it. This should reduce the level of broadcast traffic within an ELAN and enable larger ELANs to be built.

LANE 1.0 did not support multiple servers, giving single points of failure in the network. LANE 2.0 circumvents this restriction by using the LANE network-to-network interface (LNNI). This is a new part of LANE that provides a standard means to distribute LANE server components. LNNI 2.0 describes how server components can be distributed and defines the interfaces and protocols by which they exchange control information.

The ATM Forum's goal for LANE 2.0 is to enable a single ELAN to include 2000 LANE clients and 20 LANE servers and broadcast unknown servers.

It should be noted that LANE assumes the use of ATM SVCs. Although it is possible with sufficient configuration parameters to emulate SVC functionality using PVCs and thus operate LANE over PVCs, support for PVCs is beyond the scope of the LANE specification, and unlikely to be required in practice. WAN operation of LANE will require SVCs, either directly as part of a network operator's ATM service or tunnelled through an ATM virtual path network.

Multi-protocol over ATM

Whereas LANE maps layer 2 networks to ATM, multi-protocol over ATM (MPOA) [13] provides a method for overlaying layer 3 protocols (such as IP, IPX, DECnet, etc) on to ATM. MPOA, as developed, uses the next hop resolution protocol (NHRP) and LANE.

An MPOA client (MPC) wishing to exchange traffic with another such client within the same layer 3 subnet will use LANE. Traffic between clients on different layer 3

subnets will be carried as connectionless traffic over a routed network between ELANs. If the traffic is suitable (e.g. a certain amount of traffic over a time period, such as a data stream), the client will send an MPOA resolution request to the MPOA server (MPS), shown as step 1 in Fig 5. The server uses NHRP [14] to determine the ATM address of the destination client using NHRP resolution request and replies (steps 2 and 4). The destination server passes the request to the destination client which can optionally return the QoS parameters for the ATM cut-through (step 3). Standard routing protocols such as OSPF are used between servers. The ATM address of the destination client is now forwarded to the source in an MPOA resolution reply (step 5) together with any QoS information. The source will now establish a direct ATM virtual circuit 'cut-through' between itself and the destination using standard ATM signalling. This allows end-to-end inter-subnet communication without the need to traverse a number of routers.

Currently only IP and IPX are supported, although support for other layer 3 protocols is planned. MPOA has the advantage that end-systems on different layer 3 subnets can obtain direct ATM connections (via SVCs), with optional QoS, across the ATM WAN. MPOA does not support multicast traffic over ATM cut-throughs. Multicast traffic is always carried on a hop-by-hop basis.

4. LAN evolution

In the last two or so years it has become apparent that a revolution was occurring in the application environment, made possible by the greater desktop connectivity

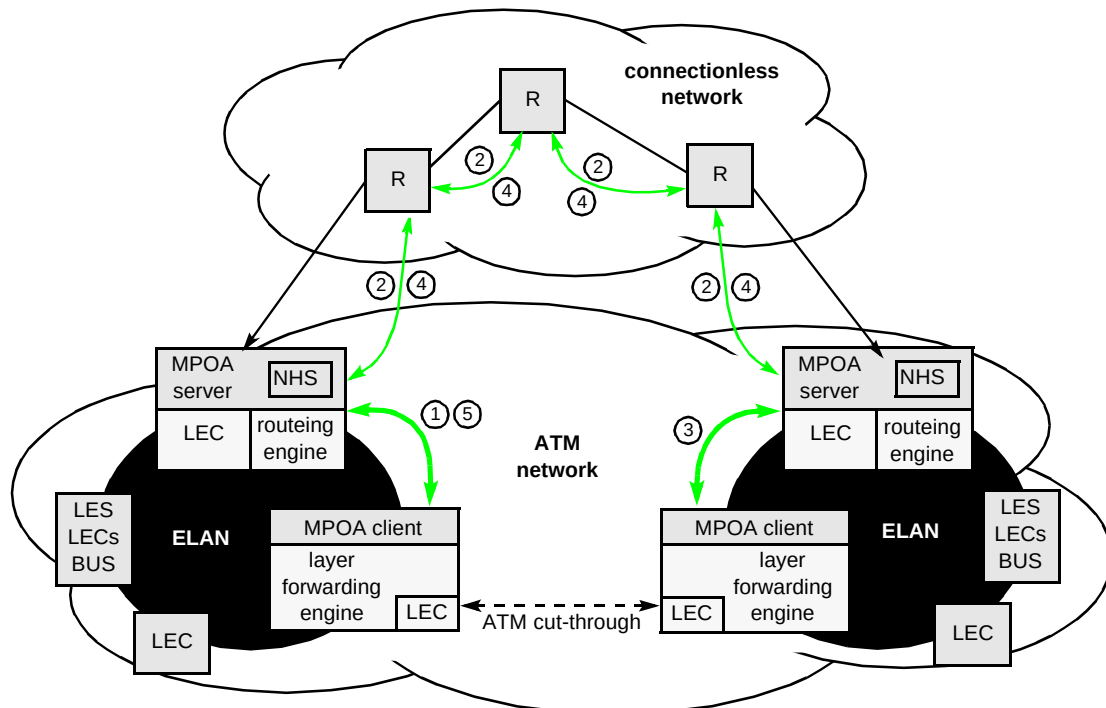


Fig 5 The establishment of an ATM cut-through in MPOA.

described above, along with the growth of the global Internet infrastructure. This revolution was the deployment of World Wide Web capability to almost every desktop in a very short period. Corporations established web servers, often with minimal correlation to individual workgroups. Employees seeking information would access a server within their building or campus, within their corporate intranet, or even across the Internet. This universal any-to-any connectivity had obvious implications on both layer 2 and layer 3 scalability. No longer could a network planner predict the amount of traffic remaining within a given workgroup — a new 80/20 rule emerged, with 80% of the traffic heading outside the workgroup making layer 3 once again of paramount importance.

A solution for layer 3 scalability in switched internetworks is thus needed. Towards the end of 1997, initial solutions in the form of distributed layer 3 switching architectures for both ATM and Ethernet were seen. In ATM, this takes the form of multi-protocol over ATM (MPOA), while for Ethernet much the same concept can be implemented, but operating on flows (a flow is defined as multiple packets between a source and destination in a given time) rather than establishing SVCs.

Under both solutions, the centrally driven function of calculating routing tables is decoupled from the process of layer 3 switching. The route calculation function is carried out at one or more sites, while the actual layer 3 switching function is distributed to the periphery (i.e. the wiring closet). A switched internetwork would now consist of LAN switching (and sometimes ATM) to the desktop, with the LAN switches possessing both layer 2 and layer 3 functionality (hence the term, multilayer switch), while ATM-connected hosts would implement MPOA. The network core would consist of ATM, fast Ethernet, and, from the beginning of 1998, Gigabit Ethernet. Finally, the route calculation function would exist on any hardware platform capable of running the classical routing protocols such as open shortest path first (OSPF), while providing the necessary manageability of inter-VLAN relationships, especially security.

Although VLANs might not be required if layer 3 forwarding performance becomes equal to that obtainable at layer 2, there are some compelling factors behind the continued deployment of VLANs. Given proper policy-based management tools, they are an effective way of providing users with access to network resources, or conversely of preventing access. Thus, VLANs would now be associated with network policy and not necessarily topology or traffic volumes. They are a method of logically segmenting the intranet, overcoming layer 2 scalability issues.

The requirement for multi-gigabit forwarding rates gives two options for future LAN backbones — ATM and Gigabit Ethernet. Both have their proponents and detractors, with most debate focusing on whether or not Gigabit Ethernet has the QoS to support applications requiring real-time QoS when loaded at high levels. The appeal of ATM is its determinism, its explicit QoS support via signalling and routing. This contrasts with the traditional datagram environment which offers QoS control via RSVP, but with no QoS-based routing protocol. Thus, any service guarantees are relative at best. Working against ATM, however, is the view of some in the field that existing solutions for the transport of data across ATM such as LANE and MPOA are overly complex. Alternatives to this protocol complexity are IP switching and multi-protocol label switching (MPLS), which includes tag switching. The first is not widely implemented and raises questions of integration within an internetwork architecture, and the second is still undergoing standardisation within the IETF. However, in the future, MPLS may be a major factor in campus ATM integration. Gigabit Ethernet is a fairly logical extension to both 10- and 100-Mbit/s Ethernet, creating a comfort factor among those familiar with implementing the technology. The one question, then, is its ability to support QoS as well as its capabilities in supporting distributed layer 3. Vendors are proposing Gigabit Ethernet switch designs capable of providing classes of service by interpreting precedence within the future IEEE802.1p standard or even capable of interworking with RSVP. At the extreme, if sufficient bandwidth is made available in the campus core, explicit QoS support will no longer be required. This may not actually be the case, especially on trunks towards the core of the network supporting large numbers of real-time flows. At a switch interface, traffic must still be served with a queuing discipline to ensure that the real-time flows receive priority.

4.1 Multi-protocol over LANs

The mechanisms used in MPOA could theoretically be extended to achieve layer 3 scalability in a LAN-switching environment. A multilayer Ethernet edge switch would be capable of querying a server for address resolution and then opening a layer 2 connection across the network. This technique is introduced as multi-protocol over LANs (MPOL), and possible operation is now described.

Consider a simple campus topology consisting of interconnected edge switches serving users on LAN segments. These users are members of one or more subnets (to use IP terminology), distributed across the campus. Traditional routers capable of layer 3 route calculation and switching are placed at one or more locations in the network. Under 100baseT VLANs or LANE (if an ATM core interconnects LAN switches), all traffic from one workgroup to another passes through the router.

Now consider a LAN switch equipped with the necessary hardware (for performance) and software which will allow it to carry out layer 3 switching. It can now forward packets from one subnet to another, by-passing the router. In order to perform this layer 3 forwarding, the level of functionality has to be determined; this is where the current approaches for MPOA and MPOL diverge. For the former, data destined for a foreign subnet (therefore following the MPOA as opposed to the LANE path) will enter the MPC (Fig 5). This MPC initiates an address resolution request (mapping an IP address to an ATM NSAP address) to its local MPS which is then passed across the MPOA domain from MPS to MPS via NHRP. Upon success of this process, the MPC will have the necessary information to open a direct ATM connection to the MPC serving the destination. The MPC in this case is not equipped with routing information; but only the required functionality to perform the necessary layer 3 transformations on the packet. Under MPOA, this cut-through connection may be initiated by the MPC or MPS depending upon traffic volumes and configuration. It should be noted that some traffic may continue to traverse the routed path.

In MPOL, the LAN switches are provided with the layer 3 functionality analogous to that within MPOA. However, in contrast to MPOA where connections are established, within MPOL flows are managed. First consider a simple topology consisting of a source and destination located on a single LAN switch, along with an attached router (Fig 6).

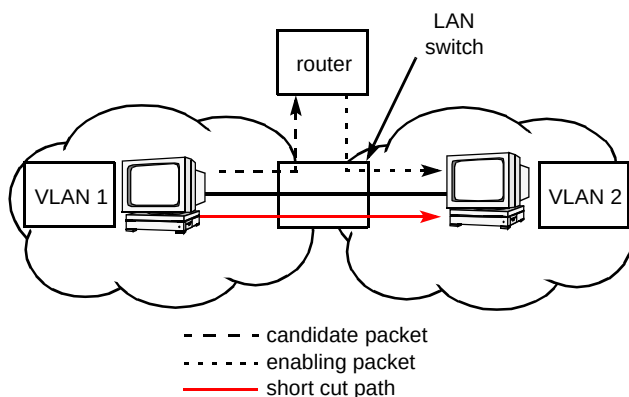


Fig 6 The simple case of forwarding packets between VLANs on a single LAN switch.

The LAN switch initially forwards all inter-subnet traffic to the router (the router's MAC address being the default for traffic leaving the subnet), but it is also capable of identifying the flow based on its layer 3 destination (and possibly other fields as well). Once the router performs the MAC rewrite, this packet is sent back through the LAN switch to the intended destination. Since the LAN switch has noted this flow, it now has the necessary information (destination layer 3 and MAC addresses) to perform a cut-through on the next packet. Under a more complex topology

with multiple LAN switches, the switch closest to the router will initially perform the cut-through, but this will shift to the LAN switch on the most efficient forwarding path between source and destination (see Fig 7). Thus, the actual layer 3 switching takes place nearest the source.

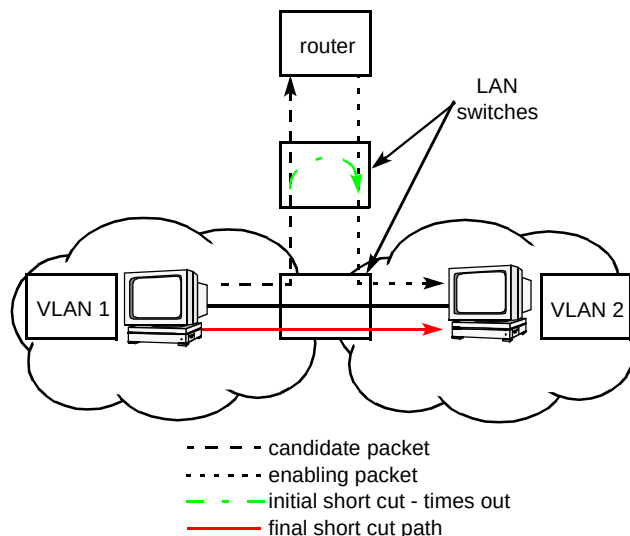


Fig 7 The concept of MPOL.

More complex topologies require deployment of an MPOA-like mechanism. Just as with MPOA, the user will control which flows are to be switched locally and which inter-subnet traffic should continue to follow the normally routed path. An important point to note is, whereas MPOA is a standard, these first approaches to MPOL will be vendor-specific, giving obvious implications for interoperability.

5. Conclusions

Switched Ethernet will be the predominant desktop delivery mechanism over LANs for the next few years. 10baseT will predominate, although 100baseT will be increasingly taken up for servers and as a LAN backboning technology — it is now economic to have 100baseT as the default networking technology in new PCs. Gigabit Ethernet, when standardised, will supplement 100baseT as a backboning technology, and may find use for attachment of servers. ATM will find use in server-farm locations, wherever QoS to the desk-top is required, and as a backbone capable of providing QoS.

While 'IP-over-Ethernet' will increasingly be the 'vanilla' option for LAN desktop delivery, the obligation to support legacy systems means that other protocols such as IPX will continue to be in widespread use, and delivery mechanisms such as Token Ring will still be in demand.

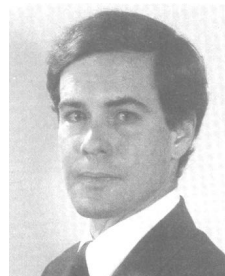
VLANs will find widespread use in campus networks, if management issues can be sensibly resolved. This may reduce the need for routers, either by properly segmenting networks (VLANs) to minimise the requirement on routers, or by the use of cut-throughs in MPOA. New techniques such as MPOL may provide means of providing wide-scale deployment of real-time applications on LAN-based networks. New multimedia applications will stimulate demand for bandwidth across the WAN.

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